ADITYA KUMAR SINGH

AI & MACHINE LEARNING ENGINEER • AUTONOMOUS SYSTEMS & RAG ARCHITECTURE

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PROFESSIONAL SUMMARY

Motivated Computer Science and Business Systems undergraduate with hands-on experience in architecting AI-powered applications utilizing **Python, FastAPI, Ollama, LangChain, Retrieval-Augmented Generation (RAG), and Docker**. Specialized in local air-gapped LLM inference, multi-agent reasoning, sub-50ms IoT UDP networking, and computer vision pipelines. Passionate about building production-grade autonomous intelligence systems.

TECHNICAL ARSENAL

Core Languages:	Python, Java, JavaScript (ES6+ Modules), SQL, HTML5, CSS3
AI & Machine Learning:	Local LLMs (Ollama, Gemma, LLaMA), RAG, LangChain, Embeddings, Multi-Agent Workflows, PyTorch, OpenCV
Backend & DevOps:	FastAPI, RESTful APIs, Docker, n8n Workflow Automation, Pinecone Vector DB, SQLite, Git, GitHub CI
Networking & Systems:	Socket Programming (UDP/TCP broadcast), Asynchronous I/O (asyncio), Web Audio API, PWA Service Workers

KEY ENGINEERING PROJECTS

AURA: Autonomous AI Voice Assistant & Tactical Station | *Python, FastAPI, Ollama, Gemma, PyAudio*

- Engineered an end-to-end autonomous desktop assistant integrating local voice synthesis (Edge-TTS), speech recognition, and OCR screen reading.
- Implemented asynchronous multi-threaded pipeline with sub-500ms voice turnarounds using Ollama local LLM inference (Gemma:2b-instruct).
- Built a real-time Tactical HUD interface displaying live system telemetry (CPU, RAM, GPU load, network state) and audio waveform feedback.

Nyaya AI: Offline Air-Gapped Legal Intelligence Assistant | *Python, Ollama, LangChain, RAG, Docker, n8n*

- Architected a 100% offline air-gapped legal research system for Indian legal statutes, case law precedents, and IPC/BNS query resolution.
- Designed a vector embedding pipeline using Ollama embeddings and Pinecone/chroma local vector store with multi-agent validation.
- Orchestrated automated statutory document ingestion pipelines via Dockerized n8n webhooks and FastAPI microservice backends.

Wiz-Bulb-Sync: Ambient IoT UDP Socket Lighting Automation | *Python, UDP Sockets, WiZ Protocol, Sub-50ms Sync*

- Developed low-latency local UDP network controller communicating directly with smart WiZ bulbs over Wi-Fi without cloud dependencies.
- Achieved sub-50ms screen color synchronization using OpenCV/Pillow frame sampling and direct JSON-over-UDP broadcast payloads.
- Created multi-light room mesh profiles, circadian rhythmic temperature transitions, and automated system tray management.

EDUCATION

B.Tech in Computer Science and Business Systems (CSBS) — **Aug 2022 – July 2026**
Techno India University • Kolkata, West Bengal — CGPA: **7.54 / 10.0**

Higher Secondary School (Class XII - Science) — **Aug 2019 – July 2021**
A. A. M. Children's Academy • Katihar, Bihar — Score: **65.8%**

Secondary School (Class X) — **Aug 2017 – May 2019**
R. K. M. Vidya Mandir • Katihar, Bihar — Score: **63.8%**

CERTIFICATIONS & ACHIEVEMENTS

- **Cybersecurity Virtual Internship** – EduSkills (Supported by Palo Alto Networks) [May – July 2023]
- **Android Developer Virtual Internship** – EduSkills (Supported by Google for Developers) [Sep – Nov 2023]
- **Networking Virtual Internship** – EduSkills (Supported by Juniper Networks) [Jan – Mar 2024]
- **Competitive Programming Profiles:** LeetCode (singhaditya7864) • GeeksforGeeks (singhaditya7864)